

Prem Madishetty

+1 (369) 210 7491 | San Diego, CA | pmadishetty7420@sdsu.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Cybersecurity Professional with 4 years of experience spanning enterprise SOC operations at NextEra Energy and advanced security research at SDSU. I specialize in bridging traditional defense with the future of the field, specifically in Post-Quantum Cryptography and the secure deployment of Generative AI. With a 3.79 GPA and certifications in AWS and Databricks, I focus on building resilient, automated architectures that protect both current infrastructure and emerging technologies.

SKILLS

- **AI & Research Security:** RAG, Agentic AI, LLM Red Teaming, NIST AI RMF & OWASP Top10 for LLMs.
- **Cryptography & Architecture:** Post-Quantum Cryptography (PQC), Kyber-512, FIPS 203, ZTA, RBAC & IAM.
- **DevSecOps & Automation:** CI/CD Pipeline, GitHub Actions, n8n, HashiCorp Vault, Kubernetes, Kyverno, Trivy & Cosign.
- **Security Operations:** Palo Alto XSOAR Cortex, CTI, CrowdStrike Falcon, Tanium, ServiceNow, Zscaler & Wireshark.
- **Cloud & Compliance:** AWS, NIST 800-53, ISO 27001/42001, GDPR, SOC 2 & HIPAA. **Languages:** Python, Bash, PowerShell & R.

EXPERIENCE

AI Security Researcher

Oct 2024 – May 2026

AI4Business Lab, San Diego State University

San Diego, CA

- Architected a "Quantum-Safe" file transfer system utilizing **PQC & Kyber-512** to mitigate "Harvest Now, Decrypt Later" risks.
- Engineered **Agentic AI** systems using **n8n** to automate peer reasoning to reduce LLM hallucinations by 14%.
- Automated LLM Red Teaming via **n8n** to stress-test models against the **OWASP Top 10** using *adversarial prompt engineering*.
- Engineered a **RAG** based CTI pipeline to ingest threat feeds to synthesize actionable **IOCs** and **TTPs** to direct incident response.
- Architected a secure *full-stack* platform with an *integrated AI chatbot* with hardened **API** endpoints and secure *web headers*.

SOC Analyst

May 2023 – July 2024

NextEra Energy

Miami, FL

- Redesigned the *Cybersecurity Awareness Mock Phishing Framework* for 9,300+ employees, reducing click rate by 30%.
- Automated incident response workflows by optimizing **SOAR** playbooks, improving operational efficiency by 40%.
- Resolved 20+ critical **Zscaler** security incidents with Wireshark to identify exfiltration attempts and restore operations.
- Authored 6 Standard Operating Procedures (**SOPs**) and a robust Business Continuity Program (**BCP**) to ensure 100% uptime.

Cybersecurity Analyst

May 2021 – April 2023

NextEra Energy

Miami, FL

- Fine-tuned **SIEM** correlation rules and **IPS/IDS** signatures, reducing false positives and accelerating MTTD by 35%.
- Analyzed 50+ malware payloads via **CrowdStrike Falcon** Sandbox to harden email filters and prevent incursions.
- Led forensics using **Tanium** and EDR to thwart 7 data exfiltration attempts, preventing reputational loss.
- Managed the **AWS IAM** rotation and **Vulnerability Management** lifecycle to ensure NIST 800-53 and ISO 27001 compliance.

TECHNICAL PROJECTS

Zero Trust Identity & Access Management (IAM) Architecture

- Architected **RBAC** for hybrid-cloud environments to minimize the internal attack surface and enforce *Principle of Least Privilege*.
- Implemented **HashiCorp Vault** for centralized *secret management* and automated rotation to eliminate hardcoded credentials.
- Engineered Adaptive Access and MFA policies using device posture and geolocation signals to enforce context-aware security.

Cloud-Native DevSecOps & Supply Chain Security

- Architected a **DevSecOps CI/CD pipeline** using *GitHub Actions* to enforce zero-trust principles in containerized workflows.
- Integrated **Trivy** and **Cosign** to automate vulnerability scanning and digital image signing, securing the software supply chain.
- Enforced Kubernetes security by implementing Kyverno admission policies to block non-compliant container images.

Enterprise AI Risk Governance & Data Sovereignty

- Orchestrated a **NIST AI RMF** risk audit of 10+ SaaS tools to identify privacy gaps and mitigate "Shadow AI" exposure.
- Architected a corporate AI Acceptable Use Policy (**AUP**) ensuring alignment with **ISO 42001** and **GDPR** data standards.
- Implemented real-time *prompt monitoring* and sensitivity labeling to prevent **PII leakage** and secure GenAI deployments.

EDUCATION

San Diego State University, San Diego, USA

August 2024 – May 2026

Master of Science in Cybersecurity Management

GPA: **3.79/4**

CERTIFICATIONS

- **CompTIA SecAI+** (In Progress, Expected May 2026)
- **AWS Cloud Fundamentals**
- **Databricks Generative AI Fundamentals**